Fatema Tuj Johora Faria

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Research Interests

- **AI Reasoning & Autonomy:** LLM Reasoning, AI Agents, Multimodal AI
- **Foundations of Dependable AI:** AI Safety, AI Security, Trustworthy AI
- **Domain-Specific AI Applications:** AI for Databases (NL-to-SQL), AI for Social Good

Education

Ahsanullah University of Science and Technology (AUST) July 2019 – December 2023
B.Sc. in Computer Science and Engineering (CGPA: 3.302 / 4.00) *Dhaka, Bangladesh*

- **Undergraduate Thesis:** Investigated generative adversarial networks (CycleGAN and Pix2Pix) for potato disease image generation to address limited training data, and evaluated the generated images for disease classification using MobileNetV2 and instance-level segmentation with Mask R-CNN.
- **Thesis Publication: IEEE CCWC 2023** and **IEEE i-COSTE 2025**.
- **Thesis Supervisor:** Prof. Dr. Mohammad Shafiul Alam & **Co-Supervisor:** Khan Md Hasib

Publications

Workshops

- **Fatema Tuj Johora Faria**, M. B. Moin, J. A. Mahmud, M. F. Mridha, and M. A. Hossain. **"CLAIR-Fin: An Adversarial Multi-Agent Framework for Claim-Level Verification and Adaptive Debate in Cross-Modal Financial QA,"** *arXiv preprint arXiv:2608.13706*, 2026. [Submitted to FinNLP-2026 @ EMNLP-2026] ↗
- **Fatema Tuj Johora Faria**, M. B. Moin, J. A. Mahmud, M. F. Mridha, and M. A. Hossain. **"LabourCrew: A Multi-Agent RAG Framework for Trustworthy Adversarial Deliberation and Statutory Reasoning over Labour Law,"** *arXiv preprint arXiv:2608.13706*, 2026. [Submitted to Natural Legal Language Processing 2026 @ EMNLP-2026] ↗
- **Fatema Tuj Johora Faria**, M. B. Moin, M. M. Rahman, K. M. Hasib, J. A. Mahmud, and M. F. Mridha. **"Guiding Language Models to Be More Empathetic: Culturally Sensitive Mental Health Advice Generation Through Human-LLM Collaboration,"** *arXiv preprint arXiv:2607.23538*, 2026. ↗

Conferences

- **Fatema Tuj Johora Faria**, M. B. Moin, M. F. Mridha, and J. A. Mahmud. **"TeachMateGPT: A Multi-Agent Knowledge-Grounded Framework for Pedagogical Assessment Generation from Science Curriculum Materials,"** *arXiv preprint arXiv:2608.13708*, 2026. [Submitted to EACL 2027] ↗
- E. Hossain, M. M. H. Nipu, **Fatema Tuj Johora Faria**, T. N. Ornee, and M. Sheikh. **"ChannelGuard: Safe Models Do Not Compose into Safe Multi-Agent Systems,"** *arXiv preprint arXiv:2607.19430*, 2026. [Submitted to SIGKDD 2027 Research Track] ↗
- **Fatema Tuj Johora Faria**, M. B. Moin, R. I. Mumu, M. M. Alam Abir, A. N. Alfy, and M. S. Alam. **"Motamot: A Dataset for Revealing the Supremacy of Large Language Models Over Transformer Models in Bengali Political Sentiment Analysis,"** *2024 IEEE Region 10 Symposium (TENSYMP)*, New Delhi, India, 2024, pp. 1–8. ↗

Journals

- **Fatema Tuj Johora Faria**, M. B. Moin, Z. Hasan, M. A. A. Khandaker, N. Islam, K. M. Hasib, and M. F. Mridha. **"MultiBanFakeDetect: Integrating Advanced Fusion Techniques for Multimodal Detection of Bangla Fake News in Under-Resourced Contexts,"** *International Journal of Information Management Data Insights*, 5(2), 100347, 2025. ↗
- **Fatema Tuj Johora Faria**, M. B. Moin, B. K. Rafa, S. Saha, M. M. Rahman, K. M. Hasib, and M. F. Mridha. **"Bangla-CalamityMMD: A Comprehensive Benchmark Dataset for Multimodal Disaster Identification in the Low-Resource Bangla Language,"** *International Journal of Disaster Risk Reduction*, 130, 105800, 2025. ↗
- **Fatema Tuj Johora Faria**, M. B. Moin, A. A. Wase, M. Ahmmed, M. R. Sani, and T. Muhammad. **"Vashantor: A Large-Scale Multilingual Benchmark Dataset for Automated Translation of Bangla Regional Dialects to Bangla Language,"** *arXiv preprint arXiv:2311.11142*, 2025. [Under Review in Language Resources and Evaluation] ↗

Professional Experience

Astha IT November 2025 – Present
AI Engineer II *Dhaka, Bangladesh*

Product: TalentX | *Python, LangGraph, LangChain, OpenAI, PostgreSQL (pgvector), WebSocket, FastAPI, React, GCP Services*

- Built a "Job Builder Agent" that combines document parsing, field-level extraction, and human-in-the-loop refinement to transform user requirements or PDF/DOCX inputs into structured, editable job specifications.
- Designed a "Profile Screening Agent" that performs resume information extraction, career-page parsing, semantic candidate–job matching, and ranking to identify qualified candidates against job-specific requirements.
- Developed a "Job Fit Scoring Agent" that dynamically generates evaluation criteria and weights from job requirements, applies weighted scoring and requirement-level candidate assessment, and sends HRs detailed evaluation reports with interpretable scores and evidence-based explanations via email.
- Architected a "Talent Matching Agent" that converts HR queries into structured retrieval constraints and combines semantic vector search, metadata filtering, query optimization, and relevance reranking to retrieve evidence-backed candidate recommendations, with dynamic removal of outdated profiles and ingestion and updating of new profiles.
- Optimized large-scale CV processing through batch execution, parallel processing, and redundant LLM-call reduction, enabling evaluation of 1,000–1,200 candidate profiles and **cutting manual effort from 5 days to under 4 hours**.

Dexian (Bangladesh) Limited July 2025 – October 2025
Senior Application Developer *Dhaka, Bangladesh*

Project: ShareFlow Agent | *Python, LlamaIndex, Azure OpenAI, Microsoft Azure Services, FastAPI, RAGAS, TypeScript, React*

- Architected a multi-agent RAG system integrating SharePoint and web sources via dynamic tool invocation, leveraging ReAct reasoning and source-aware context construction for knowledge-grounded responses.
- Developed a "Retrieval Agent" with ReAct planning, query reformulation, retrieval filtering, and custom guardrails to constrain responses to internal documents and reduce unsupported generation.
- Orchestrated an "OCR Agent" to extract content from diverse unstructured sources (scanned PDFs, images, and DOCX files) with rate limiting to support fault-tolerant performance.
- Integrated a "Web Browser Agent" with controlled web retrieval and instruction-aware source selection to augment knowledge-grounded responses with current external information when enterprise sources were insufficient.
- Implemented dynamic knowledge-base management with incremental SharePoint synchronization, file-level tracking, and memory updates, allowing users to add or remove knowledge sources without rebuilding the complete index.
- Designed session-aware conversational workflows with persistent history, context management, and session reset mechanisms, and integrated the system with Microsoft Bot Framework for deployment within Microsoft Teams.
- Directed system architecture and mentored junior developers through design reviews, debugging, code reviews, and technical knowledge-sharing, achieving an **average 60–65% annual operational cost reduction** through agent and inference optimization for 80 Sales Managers with 50+ interactions per day.

Dexian (Bangladesh) Limited May 2024 – June 2025
Application Developer *Dhaka, Bangladesh*

Project: Org Info | *Python, LlamaIndex, Azure OpenAI, Microsoft Azure Services, React, FastAPI, OpenCV, WebSocket*

- Implemented an "OrgChart Extraction Agent" with Tree-of-Thought (ToT) reasoning, visual parsing, hierarchical inference, and structured entity mapping to reconstruct organizational hierarchies from organograms and map extracted roles to corresponding Bullhorn records in a relational database.
- Developed an "OrgQuery Agent" that converts natural language queries into optimized SQL (text-to-SQL) using Chain-of-Thought (CoT) and self-consistency prompting, retrieves organizational hierarchy from the relational database, and presents results through the OrgChart interface for hierarchical visualization.
- Engineered "OrgInfo Assistant" to allow users to interact with specific organizational hierarchies by converting queries to SQL, executing them on a temporary organization-specific database to reduce load on the primary database, and converting the retrieved results to natural language.
- Created an "OrgActivity Summarization Agent" to generate 7-day summaries of organizational activities by extracting relevant notes on placements, submissions, and communication logs from the Bullhorn database.
- Automated Bullhorn synchronization through scheduled data ingestion and incremental database updates, **cutting organizational hierarchy search time by 75%** by eliminating repeated direct queries against the source system.

Project: RFPMatcher | *Python, LlamaIndex, Azure OpenAI, Microsoft Azure Services, CouchDB, React, FastAPI, WebSocket*

- Architected a document-centric RAG pipeline with Chain-of-Thought reasoning and 12 specialized analytical modules to extract key information (e.g., project scope and deliverables) from Request for Proposal (RFP) documents.
- Orchestrated a master RFP database by processing historical responses with metadata (generated answers, summaries, win/loss labels, and question categories) and vector embeddings to support semantic search.
- Designed a "Past Experience Matcher" system that analyzes new RFP documents by extracting key client requirements, generating binary (Yes/No) and descriptive questions for each section, and querying a master database to identify relevant past experiences through semantic similarity matching.
- Developed a Table of Contents (TOC) generation workflow that analyzes extracted RFP key information against pre-defined section libraries to intelligently select and prioritize relevant sections based on project requirements.
- Implemented a "Proposal Writing Assistant" that generates initial content for each TOC section using extracted RFP information and relevant past experience, enabling iterative refinement through targeted chat interactions.
- Accelerated decision-making through predictive insights, with **manual review of 100+ pages reduced by 3–5 days** and Proposal Managers able to focus on strategic bid development.

Academic Reviewing

Academic Peer Reviewer 2025 – Present

Journals & Workshops

- **Journals:** Scientific Reports; Language Resources and Evaluation; Discover Mental Health; Cluster Computing
- **Workshop:** FinNLP-2026 @ EMNLP-2026

Ongoing Research Projects

- HEAL: A Self-Correcting Multi-Agent Framework with Mutual Consistency Validation for Text-to-SQL over Clinical Patient Records
- Script, Mixing, or Switching? Disentangling the Causes of Linguistic-Form Fairness Gaps in LLMs

Technical Skills

- **Programming Languages:** Python, Java, C++
- **Web Development:** JavaScript, TypeScript, React, Tailwind CSS, FastAPI
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Deep Learning Frameworks:** TensorFlow, Keras, PyTorch
- **LLM Application Frameworks:** LangChain, LangGraph, LlamaIndex
- **Cloud Services:** Azure (Azure OpenAI, Azure SQL Database, Azure Blob Storage, Azure App Service); AWS (Amazon ECR, Amazon EC2, Amazon S3); GCP (Google Cloud Storage, Google App Engine, Google Compute Engine)
- **Others:** AlloyDB for PostgreSQL (pgvector), Milvus, Docker, Microsoft Bot Framework, OpenCV, WebSocket

Awards & Achievements

- Received a **50% registration fee scholarship** for research excellence and presentation of "*Generative Adversarial Networks for Potato Disease Classification and Instance Segmentation*" at the 11th IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2025).
- Secured a **Top 5 position among 50 multidisciplinary posters** at the AUST Research Symposium 2023: An Intra-AUST Research Exhibition for undergraduate thesis research on "*Generative Adversarial Networks for Potato Disease Classification and Instance Segmentation*".
- Recognized by **DEXIAN (Bangladesh) Limited** for **Excellence in Strategic Leadership, Agile Project Delivery, and Results-Driven Execution** in H1 2025.

Tutoring Experience

Private Academic Tutor 2020 – 2023

Mathematics, Physics & Chemistry, Grades 11–12 *Dhaka, Bangladesh*

- Provided one-on-one instruction in conceptual understanding, problem-solving, and examination preparation.
- Created customized lesson plans and practice materials aligned with individual learning needs, supported by regular assessment and constructive feedback.